Mikhail Razakov

Quantitative Research Candidate | C++ · Python

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Research-oriented CS student with professional compiler/static-analysis experience and hands-on C++/Python. I design controlled experiments, exact/reference oracles, randomized/metamorphic checks, and reproducible counterexample studies; my strongest current evidence is in algorithms, program analysis, and systems research.

Controlled experiments

UniversalToolchain: frozen corpus, valid controls, post-freeze holdouts, paired comparison, and repeated timing runs.

Mathematical & algorithmic reasoning

PS-form: normalization, modular/GCD reasoning, exact affine analysis, and an independent exact oracle.

C++ / Python research tooling

LLVM, Python harnesses, randomized/metamorphic testing, reproducible counterexamples, and differential execution.

EXPERIENCE

2026 — present

ISP RAS — Static Analysis Engineer

- Contribute to SharpChecker, ISP RAS's industrial static-analysis platform for C#/.NET.

July 1 — August 31, 2026

MCST — Compiler Engineering Intern

- Implemented an LLVM LICM pass in C++; transformation legality is derived from loop structure, memory accesses, side effects, and speculative-safety constraints.
- Built a Python harness comparing execution before/after optimization and checking explicit transformation / no-transformation cases; also implemented RPO, Dijkstra, Dinic, and Tarjan SCC in C++23.

PROJECTS

UniversalToolchain

Current exact test manifest: 1,324/1,324. In a frozen-corpus verifier experiment, B2 detected 32/32 primary and 10/10 challenge operators with 0/100 false positives on valid controls; four post-freeze holdouts were checked separately.

PS-form Memory Dependence Analyzer

Conservative yes/no/maybe results checked by an independent exact oracle on small domains, randomized/metamorphic tests, and reproducible counterexamples.

LLVM Interprocedural Global IV Optimization

A conservative LLVM 22 prototype/MVP with an actual IR transform; 29 positive/negative regression cases cover verifier validity, idempotence, and before/after execution.

SKILLS

Programming
C++23, Python, C17, C#

Experimental methods
controlled experiments, exact/reference oracles, randomized/metamorphic testing, reproducibility

Algorithms & CS
data structures, graph algorithms, program analysis, CFG/SSA, compiler optimizations

Research tooling
Python harnesses, Linux, Git, CMake, counterexample analysis

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

Prize-winner in HSE Vysshaya Proba in competitive and industrial programming.

Moscow / remote